# MSG No. 53.331 $P_{b}mna$ [ Type IV, orthorhombic ]

Table 1: Wyckoff site: `4a`, site symmetry: `2/m..`

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\ 0,\ 0]$ | `[1,2,5,6]` |
| 2 | $\left[\frac{1}{2},\ 0,\ \frac{1}{2}\right]$ | `[3,4,7,8]` |
| 3 | $\left[0,\ \frac{1}{2},\ 0\right]$ | `[9,10,13,14]` |
| 4 | $\left[\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$ | `[11,12,15,16]` |

Table 2: Wyckoff site: `4b`, site symmetry: `2/m..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2},\ 0,\ 0\right]$ | `[1,2,5,6]` |
| 2 | $\left[0,\ 0,\ \frac{1}{2}\right]$ | `[3,4,7,8]` |
| 3 | $\left[\frac{1}{2},\ \frac{1}{2},\ 0\right]$ | `[9,10,13,14]` |
| 4 | $\left[0,\ \frac{1}{2},\ \frac{1}{2}\right]$ | `[11,12,15,16]` |

Table 3: Wyckoff site: `4c`, site symmetry: `2'/m..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{2},\ \frac{1}{4},\ 0\right]$ | `[1,6,10,13]` |
| 2 | $\left[\frac{1}{2},\ \frac{3}{4},\ 0\right]$ | `[2,5,9,14]` |
| 3 | $\left[0,\ \frac{1}{4},\ \frac{1}{2}\right]$ | `[3,8,12,15]` |
| 4 | $\left[0,\ \frac{3}{4},\ \frac{1}{2}\right]$ | `[4,7,11,16]` |

Table 4: Wyckoff site: `4d`, site symmetry: `2'/m..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[0,\ \frac{1}{4},\ 0\right]$ | `[1,6,10,13]` |
| 2 | $\left[0,\ \frac{3}{4},\ 0\right]$ | `[2,5,9,14]` |
| 3 | $\left[\frac{1}{2},\ \frac{1}{4},\ \frac{1}{2}\right]$ | `[3,8,12,15]` |
| 4 | $\left[\frac{1}{2},\ \frac{3}{4},\ \frac{1}{2}\right]$ | `[4,7,11,16]` |

Table 5: Wyckoff site: `8e`, site symmetry: `2..`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ 0,\ 0]$ | `[1,2]` |
| 2 | $\left[\frac{1}{2}-x,\ 0,\ \frac{1}{2}\right]$ | `[3,4]` |
| 3 | $[-x,\ 0,\ 0]$ | `[5,6]` |
| 4 | $\left[x+\frac{1}{2},\ 0,\ \frac{1}{2}\right]$ | `[7,8]` |
| 5 | $\left[x,\ \frac{1}{2},\ 0\right]$ | `[9,10]` |
| 6 | $\left[\frac{1}{2}-x,\ \frac{1}{2},\ \frac{1}{2}\right]$ | `[11,12]` |
| 7 | $\left[-x,\ \frac{1}{2},\ 0\right]$ | `[13,14]` |

| | | |
|---|---|---|
| 8 | $\left[x+\frac{1}{2},\ \frac{1}{2},\ \frac{1}{2}\right]$ | `[15,16]` |

Table 6: Wyckoff site: `8f`, site symmetry: `2’..`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[x,\ \frac{1}{4},\ 0\right]$ | `[1,10]` |
| 2 | $\left[x,\ \frac{3}{4},\ 0\right]$ | `[2,9]` |
| 3 | $\left[\frac{1}{2}-x,\ \frac{1}{4},\ \frac{1}{2}\right]$ | `[3,12]` |
| 4 | $\left[\frac{1}{2}-x,\ \frac{3}{4},\ \frac{1}{2}\right]$ | `[4,11]` |
| 5 | $\left[-x,\ \frac{3}{4},\ 0\right]$ | `[5,14]` |
| 6 | $\left[-x,\ \frac{1}{4},\ 0\right]$ | `[6,13]` |
| 7 | $\left[x+\frac{1}{2},\ \frac{3}{4},\ \frac{1}{2}\right]$ | `[7,16]` |
| 8 | $\left[x+\frac{1}{2},\ \frac{1}{4},\ \frac{1}{2}\right]$ | `[8,15]` |

Table 7: Wyckoff site: `8g`, site symmetry: `.2.`

| No. | position | mapping |
|---|---|---|
| 1 | $\left[\frac{1}{4},\ y,\ \frac{1}{4}\right]$ | `[1,3]` |
| 2 | $\left[\frac{1}{4},\ -y,\ \frac{3}{4}\right]$ | `[2,4]` |
| 3 | $\left[\frac{3}{4},\ -y,\ \frac{3}{4}\right]$ | `[5,7]` |
| 4 | $\left[\frac{3}{4},\ y,\ \frac{1}{4}\right]$ | `[6,8]` |
| 5 | $\left[\frac{1}{4},\ y+\frac{1}{2},\ \frac{1}{4}\right]$ | `[9,11]` |
| 6 | $\left[\frac{1}{4},\ \frac{1}{2}-y,\ \frac{3}{4}\right]$ | `[10,12]` |
| 7 | $\left[\frac{3}{4},\ \frac{1}{2}-y,\ \frac{3}{4}\right]$ | `[13,15]` |
| 8 | $\left[\frac{3}{4},\ y+\frac{1}{2},\ \frac{1}{4}\right]$ | `[14,16]` |

Table 8: Wyckoff site: `8h`, site symmetry: `m..`

| No. | position | mapping |
|---|---|---|
| 1 | $[0,\ y,\ z]$ | `[1,6]` |
| 2 | $[0,\ -y,\ -z]$ | `[2,5]` |
| 3 | $\left[\frac{1}{2},\ y,\ \frac{1}{2}-z\right]$ | `[3,8]` |
| 4 | $\left[\frac{1}{2},\ -y,\ z+\frac{1}{2}\right]$ | `[4,7]` |
| 5 | $\left[0,\ y+\frac{1}{2},\ z\right]$ | `[9,14]` |
| 6 | $\left[0,\ \frac{1}{2}-y,\ -z\right]$ | `[10,13]` |
| 7 | $\left[\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$ | `[11,16]` |
| 8 | $\left[\frac{1}{2},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$ | `[12,15]` |

Table 9: Wyckoff site: `16i`, site symmetry: `1`

| No. | position | mapping |
|---|---|---|
| 1 | $[x,\ y,\ z]$ | `[1]` |
| 2 | $[x,\ -y,\ -z]$ | `[2]` |
| 3 | $\left[\frac{1}{2}-x,\ y,\ \frac{1}{2}-z\right]$ | `[3]` |

*continued ...*

Table 9

| No. | position | mapping |
|---|---|---|
| 4 | $\left[\frac{1}{2}-x,\ -y,\ z+\frac{1}{2}\right]$ | [4] |
| 5 | $[-x,\ -y,\ -z]$ | [5] |
| 6 | $[-x,\ y,\ z]$ | [6] |
| 7 | $\left[x+\frac{1}{2},\ -y,\ z+\frac{1}{2}\right]$ | [7] |
| 8 | $\left[x+\frac{1}{2},\ y,\ \frac{1}{2}-z\right]$ | [8] |
| 9 | $\left[x,\ y+\frac{1}{2},\ z\right]$ | [9] |
| 10 | $\left[x,\ \frac{1}{2}-y,\ -z\right]$ | [10] |
| 11 | $\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$ | [11] |
| 12 | $\left[\frac{1}{2}-x,\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$ | [12] |
| 13 | $\left[-x,\ \frac{1}{2}-y,\ -z\right]$ | [13] |
| 14 | $\left[-x,\ y+\frac{1}{2},\ z\right]$ | [14] |
| 15 | $\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ z+\frac{1}{2}\right]$ | [15] |
| 16 | $\left[x+\frac{1}{2},\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$ | [16] |